

Part 2

{ Mid 2023 }

Questions (16,17)

The eddy-current loss in a certain electrical apparatus operating at its rated voltage and rated frequency of 240-V and 25-Hz is 642-W. If the apparatus is connected to a 60 Hz source, whose voltage is such as to cause the flux density to be 62% of its rated value.

16. The eddy current loss in this case is equal to

- A. 1-kW ~~B. 1.42-kW~~ C. 0.5-kW D. 3-kW

17. The value of applied voltage with 60 Hz to maintain the same flux density of 240 V, 25 Hz, is equal to

- ~~A. 576-V~~ B. 240-V C. 357-V D. 280-V

Questions (18,19)

In a 440-V, 50-Hz core, the total core loss is 2500-W. When the applied voltage is 220-V at 25-Hz the corresponding loss is 850-W.

18. The eddy current loss at 440-V and 50-Hz is

- ~~A. 900-W~~ B. 1200-W ~~C. 1600-W~~ ~~D. 1000-W~~

19. The hysteresis loss at 440-V and 50-Hz is

- ~~A. 1600-W~~ B. 1000-W C. 1200-W ~~D. 900-W~~

20. Open circuit test of transformer should be done at ~~rated voltage~~ D. all mentioned
A. reduced voltage B. double voltage

21. The main purpose of performing short circuit test on a transformer is to determine its ~~resistance~~ D. B & C
A. insulation B. copper loss C. R_{eq} and X_{eq}

22. The main purpose of performing a back-to-back test in a transformer is to determine
~~A. The maximum rise temperature at full load~~ B. The no load losses
C. The full load copper losses D. all mentioned

23. Consider the two-winding transformer of 110 V-55 V. For the polarity test, if the transformer has additive polarities, so the reading of test voltmeter is
A. 55-V B. 0-V C. 110-V ~~D. 165-V~~

Questions (24-27)

50 kVA, 440-220 V, 50 Hz, single phase, two winding, step down transformer, has an equivalent impedance of $0.1596 \angle 76.9^\circ \Omega$ referred to high voltage side, if the transformer is feeding a load at rated voltage, 80% of full load current and 0.7 pf lagging. The transformer core loss is 85% of full load copper losses.

24. The value of input voltage is equal to

- A. 440-V ~~B. 452.1-V~~ C. 354.54-V D. 225-V

25. The percentage voltage regulation is equal to

- A. 2.83% ~~B. 2.73%~~ C. 0% D. 2.28%

26. The efficiency of transformer at this load is equal to

- ~~A. 100%~~ B. 85% ~~C. 97.5%~~ D. 87.2%

27. The maximum efficiency is occurring at percentage of full load equal to

- ~~A. 92.2%~~ B. 100% C. 97.3% D. 50%

28. The percent of sharing each transformer to total bank-current drawn by the load is approximately

A. $I_A = 0.698 I_{bank}$, $I_B = 0.307 I_{bank}$

B. $I_A = 0.5 I_{bank}$, $I_B = 0.5 I_{bank}$

☒ C. $I_A = 0.307 I_{bank}$, $I_B = 0.698 I_{bank}$

D. $I_A = 0.75 I_{bank}$, $I_B = 0.25 I_{bank}$

29. The maximum load that can be handled by the bank without overloading any one transformers is

A. $S_{bank} = 520\text{-kVA}$

☒ B. $S_{bank} = 244\text{-kVA}$

C. $S_{bank} = 275\text{-kVA}$

D. $S_{bank} = 175\text{-kVA}$

30. At this load, the performance of TA, and TB will be as follow:

A. TA is full load, and high power factor

☒ B. TB is full load and low power factor

C. TB is not full load, but low power factor

☒ D. A & C

Questions (28 → 30)

Two Transformers with 2400-240 V

Tr. A	Tr. B
75 KVA	200 KVA
$Z_{Aeq_H} = 2.734 \angle 62.5^\circ$	$Z_{Beq_H} = 1.2031 \angle 74.7^\circ$

Solution

$$P_{e1} = 642 \text{ W} \quad V_1 = 240 \text{ V} \quad f_1 = 25 \text{ Hz}$$

$$B_2 = 0.62 B_1 \quad P_{e2} = ? \quad f_2 = 60 \text{ Hz}$$

16) $P_e \propto B^2 f^2$

$$\frac{P_{e1}}{P_{e2}} = \left(\frac{B_1}{B_2} \right)^2 \left(\frac{f_1}{f_2} \right)^2$$

$$\frac{642}{P_{e2}} = \left(\frac{B_1}{0.62 B_1} \right)^2 \left(\frac{25}{60} \right)^2$$

$$\therefore P_{e2} = 1.42 \text{ kW}$$

17) $\frac{V}{f} = \text{const} \rightarrow B = \text{const}$

$$\frac{240}{25} = \frac{V_2}{60}$$

$$\therefore V_2 = 576 \text{ V}$$

$$V_1 = 440 \text{ V} \quad F_1 = 50 \text{ Hz} \quad P_{C1} = 2500 \text{ W}$$

$$V_2 = 220 \text{ V} \quad F_2 = 25 \text{ Hz} \quad P_{C2} = 850 \text{ W}$$

$$P_e = ? \quad P_h = ? \quad \text{at } \begin{cases} V = 440 \text{ V} \\ F = 50 \text{ Hz} \end{cases}$$

$$18) \quad \frac{V}{F} = \text{const} \rightarrow B = \text{const}$$

$$P_C = P_e + P_h = K_e' f^2 + K_h' f$$

$$2500 = K_e' (50)^2 + K_h' (50) \rightarrow \textcircled{1}$$

$$850 = K_e' (25)^2 + K_h' (25) \rightarrow \textcircled{2}$$

$$\therefore K_e' = 0.64$$

$$K_h' = 18$$

$$\therefore P_e = 0.64 * 50^2 = 1600 \text{ W}$$

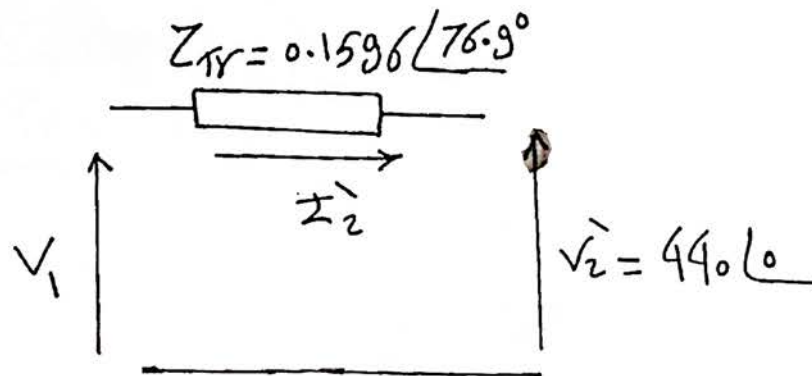
$$19) \quad \therefore P_h = 18 * 50 = 900 \text{ W}$$

20) C

21) C

22) A

23) Additive $\rightarrow V = 110 + 55 = 165 \text{ V}$



$$I_{F.L} = \frac{60,000}{440} \angle -45.57^\circ$$

$$I_2 = 0.8 I_{F.L}$$

24) $V_1 = V_2 + I_2 \cdot Z_{TY}$

$$\therefore V_1 = 452.4 \angle 0.955^\circ$$

$$1) V.R = \frac{V_1 - V_2}{V_2} \times 100\%$$

$$\therefore V.R = 2.73\%$$

$$26) Z_{TV} = 0.036 + j0.155 \Omega$$

$$P_{Cu f.L} = (I_{2 f.L})^2 R_{eq} = 464.876 \text{ W}$$

$$\eta = \frac{0.8 \times 50,000 \times 0.7}{0.8 \times 50,000 \times 0.7 + 0.85 P_{Cu} + (0.8)^2 P_{Cu f.L}} \times 100\%$$

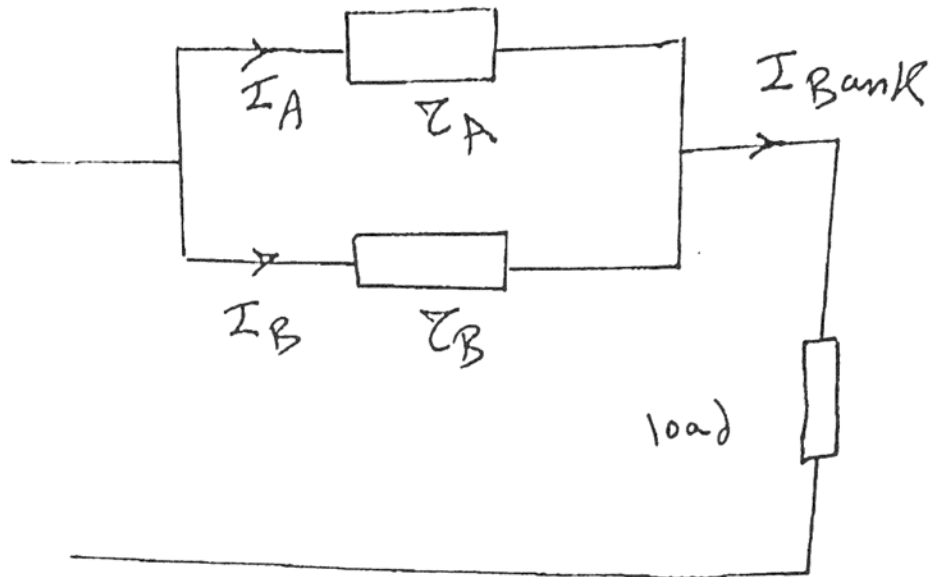
$$\therefore \eta = 97.5\%$$

$$27) S_{\eta_{max}} = S_{f.L} \sqrt{\frac{P_C}{P_{Cu f.L}}}$$

$$\frac{S_{\eta_{max}}}{S_{f.L}} = \sqrt{\frac{0.85 P_{Cu f.L}}{P_{Cu f.L}}} = \sqrt{0.85}$$

$$\therefore \frac{S_{\eta_{max}}}{S_{f.L}} = 92.2\%$$

28)



$$I_A = I_{Bank} * \left| \frac{Z_B}{Z_B + Z_A} \right|$$

$$\therefore I_A = 0.307 I_{Bank}$$

$$I_B = I_{Bank} * \left| \frac{Z_A}{Z_A + Z_B} \right|$$

$$\therefore I_B = 0.698 I_{Bank}$$

$$I_{A \text{ rated}} = \frac{S_A}{2400} = 31.25 \text{ A}$$

$$I_{B \text{ rated}} = \frac{S_B}{2400} = 83.33 \text{ A}$$

Case.1

$$I_B = I_{B \text{ rated}} = 83.33 \text{ A}$$

$$\therefore I_{\text{Bank}} = \frac{83.33}{0.698} = 119.389 \text{ A}$$

$$\therefore I_A = 0.307 I_{\text{Bank}} = 36.65$$

$$I_A > I_{A \text{ rated}} \rightarrow \text{overloaded}$$

$\therefore$ Refused case

Case.2

$$I_A = I_{A \text{ rated}} = 31.25 \text{ A}$$

$$I_{\text{Bank}} = 101.7915 \text{ A}$$

$$I_B = 71.05 \text{ A}$$

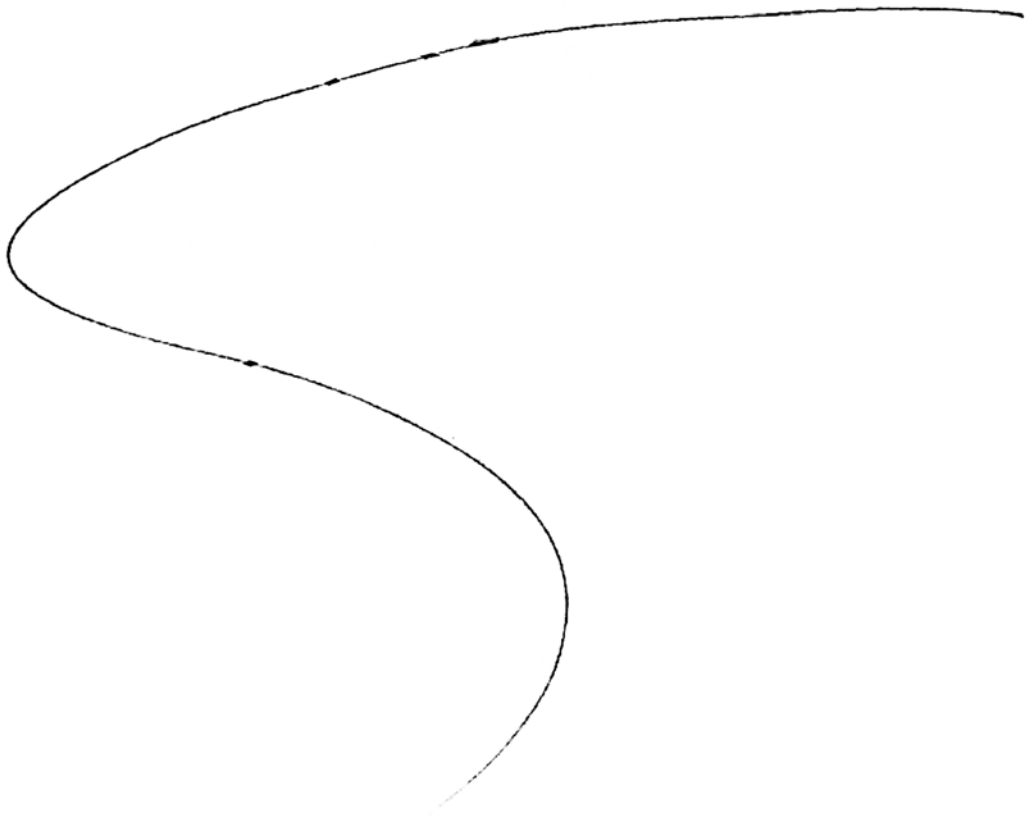
$$I_B < I_{B \text{ rated}}$$

$\therefore$ No overload

$$S_{\text{Bank}} = \sqrt{3} I_{\text{Bank}} = 244 \text{ KVA}$$

30) A & C

Note loading $\uparrow$: PF $\uparrow$



Midterm 2022 Part.2

Questions (16→22)

- A single phase two winding transformer has 500-turns in its primary winding. When it is connected to a single-phase 120-V, 60-Hz power supply, the no load current is 1.6-A, and the no load power is 80-W. neglect the winding resistance and leakage reactance of the winding. Calculate:
- 16) The core loss current is
a. ~~2/3-A~~ b. 1.5-A c. 1.45-A d. 3-A
- 17) The magnetizing current is
a. 2/3-A b. 1.5-A c. ~~1.45-A~~ d. 3-A
- 18) The peak value of the core flux is
a. 0.2-mWb b. ~~0.9-mWb~~ c. 0.5-mWb d. 1.2-mWb
- 19) The magnetizing reactance X_m ,
a. ~~82.76-Ω~~ b. 180-Ω c. 60-Ω d. 75-Ω
- 20) The core loss resistance R_c
a. 82.76-Ω b. ~~180-Ω~~ c. 60-Ω d. 75-Ω
- 21) Eddy current losses in core is measured at $f = 50$ -Hz, it is found to be 200-W, at the same flux density and frequency of 60-Hz, it will equal to
a. 180-W b. 242-W c. 200-W d. ~~288-W~~
- 22) Hysteresis losses in core is measured at $f = 50$ -Hz, it is found to be 300-W, at the same flux density and frequency of 60-Hz, it will equal to
a. ~~360-W~~ b. 180-W c. 540-W d. 432-W

• Questions (23→25)

- A single-phase, 10-kVA, 460/120 V, 50-Hz transformer has a maximum efficiency of 96 % when it delivers 80% of rated load at 0.8 power factor lagging.

- 23) The copper losses at full load is
a. 0.1333-kW b. ~~0.208-kW~~ c. 0.2666-kW d. 0.416-kW
- 24) The total core losses at rated voltage is
a. ~~0.1333-kW~~ b. 0.208-kW c. 0.2666-kW d. 0.416-kW
- 25) The efficiency at full load and 0.75 pf leading is
a. 50% b. 97.6% c. ~~95.6%~~ d. 92.5%

26) For parallel operation of two transformers, they should have

- a. The same turns ratio and voltage rating
- b. The same % Impedance, and X/R ratio
- c. The same polarity
- ☒ d. All of the above

27) If two transformers connected in parallel with different X/R ratio

- a. Each transformer will share different power factor
- b. Each transformer will share the same load's power factor
- c. One transformer could be over loaded
- ☒ d. (a), and (c)

• Questions (28→30)

• Two transformers have a turns ratio equivalent to their 2400 - 240 volta

The data of each one is listed in the following table:

A	B
75-kVA	200-kVA
$Z_{Aeq} = 2.7342 \angle 62.571^\circ \Omega$	$Z_{Beq} = 1.2031 \angle 74.733^\circ \Omega$

28) The percent of the total bank -current drawn by each transformer is

a. $I_A = 0.698 I_{bank}$, $I_B = 0.307 I_{bank}$

b. $I_A = 0.5 I_{bank}$, $I_B = 0.5 I_{bank}$

☒ c. $I_A = 0.307 I_{bank}$, $I_B = 0.698 I_{bank}$

d. $I_A = 0.75 I_{bank}$, $I_B = 0.25 I_{bank}$

29) The maximum load that can be handled by the bank without overloading of the transformers

a. $S_{bank} = 520\text{-kVA}$

☒ b. $S_{bank} = 244\text{-kVA}$

c. $S_{bank} = 275\text{-kVA}$

d. $S_{bank} = 175\text{-kVA}$

30) The % voltage regulation, if the maximum load is unity power factor

a. 1.8 %

b. -3.2 %

☒ c. 1.2 %

Solution

$$N_1 = 500 \text{ Turns}$$

$$V_1 = 120 \text{ V}$$

$$f = 60 \text{ Hz}$$

$$I_0 = 1.6 \text{ A}$$

$$P_0 = 80 \text{ W}$$

$$16) \quad I_c = I_0 \cos \Phi_0$$

$$\cos \Phi_0 = \frac{P_0}{V_0 I_0} = \frac{80}{120 \times 1.6} = 0.4167 \#$$

$$\therefore I_c = 1.6 \times 0.4167 = 0.667 \text{ A} \#$$

$$17) \quad I_m = I_0 \sin \Phi_0 = 1.6 \times \sin(\cos^{-1}(0.4167))$$

$$\therefore I_m = 1.45 \text{ A} \#$$

$$18) \quad E = 120 = 4.44 N_1 \Phi_{\max} f$$

$$\therefore \Phi_{\max} = \frac{120}{4.44 \times 500 \times 60} = 9 \times 10^{-4} \text{ wb}$$

$$19) X_m = \frac{120}{1.45} = 82.76 \Omega \quad \equiv$$

$$20) R_c = \frac{120}{0.667} = 180 \Omega \quad \equiv$$

$$21) P_e = K_e B^2 f^2 \text{ Vol}$$

$\because B = \text{const} \rightarrow P_e \propto f^2$

$$\frac{P_{e1}}{P_{e2}} = \left(\frac{f_1}{f_2} \right)^2$$

$$\frac{200}{P_{e2}} = \left(\frac{50}{60} \right)^2$$

$$\therefore P_{e2} = 288 \text{ W} \quad \equiv$$

$$22) P_h = K_h B_{\max} f \text{ Vol}$$

$$P_h \propto f$$

$$\frac{P_{h1}}{P_{h2}} = \frac{f_1}{f_2} \rightarrow \frac{300}{P_{h2}} = \frac{50}{60}$$

$$\therefore P_{h2} = 360 \text{ W} \quad \equiv$$

1 Φ Transformer $S_{f.L} = 10 \text{ KVA}$

460/120 V, 50 Hz, $\eta_{max} = 96\%$

$S_{\eta_{max}} = 0.8 S_{f.L} = 8 \text{ KVA}$ $Pf = 0.8 \text{ lag}$

$$23) \quad \eta_{max} = \frac{S_{\eta_{max}} Pf}{S_{\eta_{max}} Pf + 2P_C} = 0.96$$
$$\frac{8000 \times 0.8}{8000 \times 0.8 + 2P_C} = 0.96$$

$$\therefore P_C = 133.33 \text{ W} \quad \#$$

$P_C = P_{Cu} = 133.33 \text{ W}$ at 80% of full load

$P_{Cu} \propto X^2$ $X \rightarrow \text{Percentage of loading}$

$$\frac{P_{Cu1}}{P_{Cu2}} = \left(\frac{X_1}{X_2} \right)^2$$

$$\frac{133.33}{P_{Cu f.L}} = \left(\frac{0.8}{1} \right)^2$$

$$\therefore P_{Cu f.L} = 208.33 \text{ W} \quad \#$$

$$24) P_C = 133.33 \text{ W}$$

note $\rightarrow P_{n.l} = P_C$ doesn't depend on loading

$$25) \eta_{f.l} = \frac{10,000 \times 0.75}{(10,000 \times 0.75) + (133.33) + 208.33} \times 100 \%$$

$$\therefore \eta_{f.l} = 95.6 \%$$

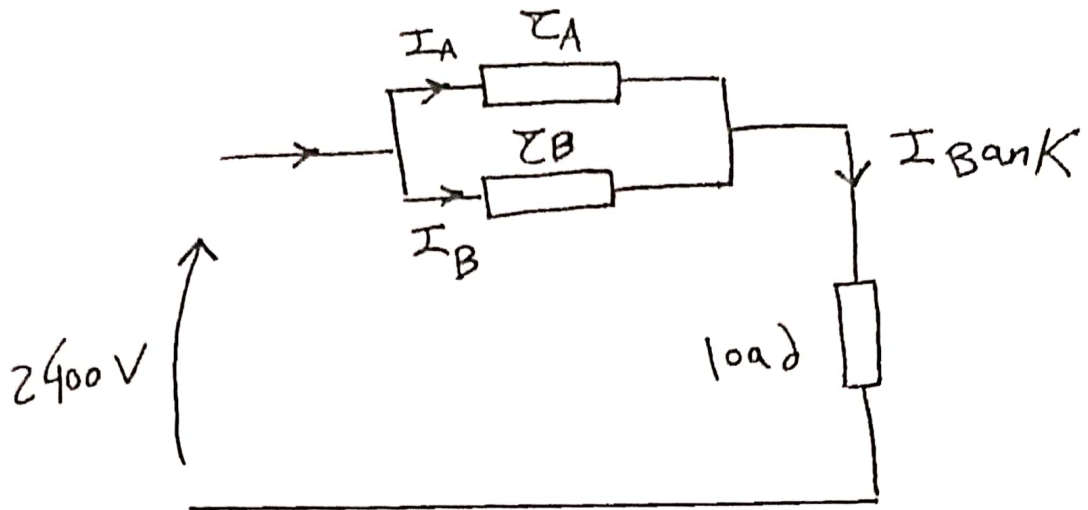
26)

d

27)

d

28)



$$I_A = I_{Bank} * \left| \frac{Z_B}{Z_A + Z_B} \right|$$

$$\therefore I_A = 0.307 I_{Bank} \quad \#$$

$$I_B = I_{Bank} - I_A = 0.698 I_{Bank} \quad \#$$

29)

$$I_{A \text{ rated}} = \frac{S_A}{V_1} = \frac{75000}{2400}$$

$$\therefore I_{A \text{ rated}} = 31.25 \text{ A}$$

$$I_{B \text{ rated}} = \frac{S_B}{V_1} = \frac{200,000}{2400}$$

$$\therefore I_{B \text{ rated}} = 83.33 \text{ A}$$

Case 1

$$\text{IF } I_B = I_{B \text{ rated}} = 83.33 \text{ A}$$

$$\therefore I_{\text{Bank}} = \frac{83.33}{0.698} = 119.389 \text{ A}$$

$$\therefore I_A = 0.307 \times 119.389 = 36.65 \text{ A}$$

$$I_A > I_{A \text{ rated}} \text{ (overloaded)}$$

$\therefore$ {Refuse}

Case 2

$$\text{IF } I_A = I_{A \text{ rated}} = 31.25 \text{ A}$$

$$\therefore I_{\text{Bank}} = 101.7915 \text{ A}$$

$$\therefore I_B = 71.05 \text{ A}$$

$$I_B < I_{B \text{ rated}}$$

$\therefore$ No over load

$$\therefore S_{\text{Bank}} = V_1 \times I_{\text{Bank}} = 244 \text{ KVA}$$

30)

$$V_1 = V_2' + I_{\text{bank}} Z_{eq}$$

$$V_1 = 2400 \angle 0^\circ + (101.79 \angle 0^\circ)(0.8 \angle 71.024^\circ)$$

$$\therefore V_1 = 2427.701 \angle 1.8^\circ$$

$$V.R = \frac{|V_1| - |V_2'|}{|V_2'|} \times 100\%$$

$$\therefore V.R \approx 1.2\%$$



DC machines and Transformers (EEP 231)
2nd level Elec. Power Program
Time allowed: 1 hour

آلات التيار المستمر والمحولات
السنة الدراسية : المستوى الثاني كهرباء قوى
مدة الإمتحان : ساعة

Choose the **BEST** answer for the following questions. Please write your answers on the **ANSWER SHEET ONLY.**

Part II: Transformers

A single-phase transformer whose primary is connected to 400V, 50Hz supply, takes a magnetizing current of 0.5 A, the power factor on no-load is 0.1 lagging. For the given data, solve from q16 to q20.

16. The no-load current is

- a. 0.05 A ☒ b. 0.5 A c. 0.005 A ☒ d. 1.5 A

17. The iron loss current is

- ☒ a. 0.05 A b. 0.5 A c. 0.005 A d. 1.5 A

18. The iron core loss power is

- a. 200 W b. 0.2 W c. 2 W ☒ d. 20 W

19. The iron core resistance is

- a. 1000 Ω ☒ b. 8000 Ω c. 800 Ω d. 16000 Ω

20. The magnetizing reactance is

- a. 1000 Ω b. 8000 Ω ☒ c. 800 Ω d. 16000 Ω

A 240/120V, 2 kVA, 50 Hz, single-phase transformer, when the low voltage side is short-circuited, the high voltage side data are 40 V, 8 A and 192 W. If the transformer works at full-load and 0.6 pf leading with load voltage of 120 V. The core loss is 40 W. Use the approximate equivalent circuit to solve from q21 to q26.

21. The equivalent resistance referred to HV side is

☒ a. 3 Ω	b. 4 Ω	c. 5 Ω	d. 24 Ω
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22. The equivalent inductance referred to HV side is

a. 4 H	b. 0.016 H	☒ c. 0.0127 H	d. 3 H
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23. The magnitude of primary voltage is

a. 240 V	b. 281.6 V	c. 267.1 V	☒ d. 231.8 V
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24. The percentage voltage regulation is

a. 3.4%	☒ b. -3.4%	c. 17.3%	d. 0%
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25. The full-load efficiency is

☒ a. 82.8%	b. 85.2%	c. 88.9%	d. 89.8%
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26. The load at which the maximum efficiency occurs is

a. 500 VA	b. 384 VA	c. 876 VA	d. 4560 VA
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In a 440 V, 50 Hz core, the total core loss is 2500 W. When the applied voltage and frequency are halved, the corresponding loss is 850 W. Solve from q27 to q30.

27. The eddy current loss at 440 V and 50 Hz is

a. 450 W	b. 900 W	☒ c. 1600 W	d. 400 W
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28. The hysteresis loss at 440 V and 50 Hz is

a. 1600	☒ b. 900 W	c. 450 W	d. 400 W
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29. The hysteresis loss at 220 V and 25 Hz is

a. 1600	b. 900 W	☒ c. 450 W	d. 400 W
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30. When the applied voltage and frequency are halved, the maximum flux density

☒ a. increases	b. decreases	☒ c. remains constant	
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Good Luck

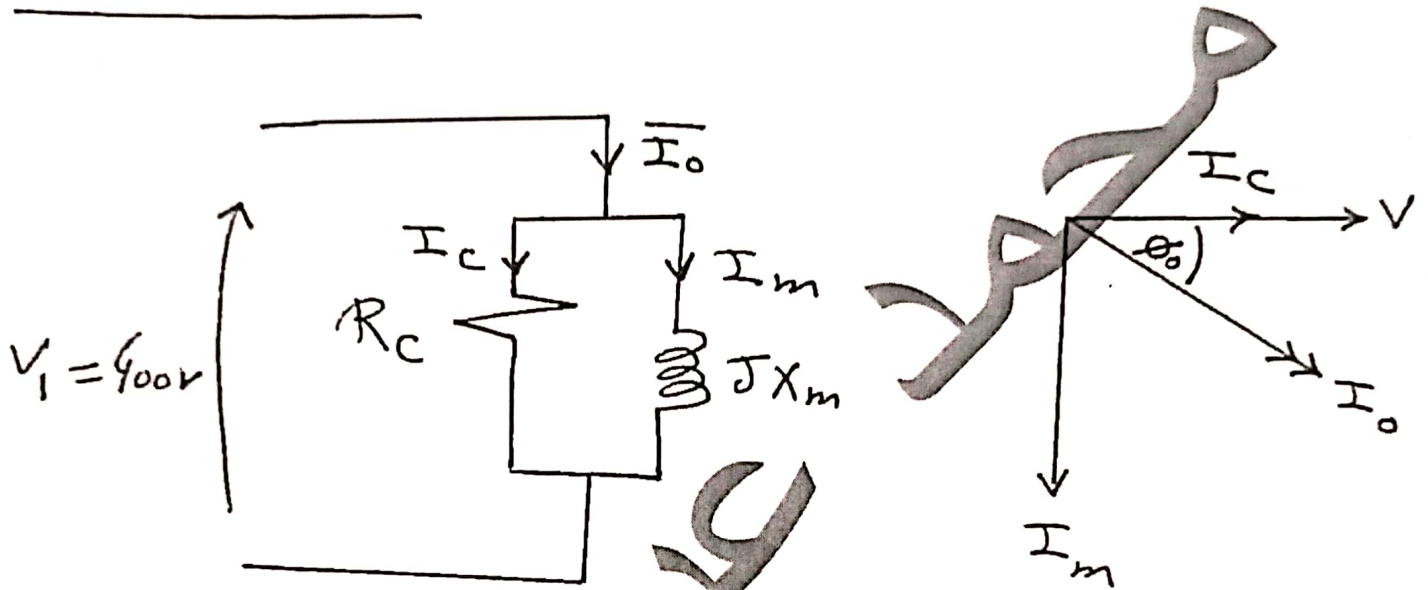
Prof. Dr. Mohamed Yousry and Dr. Hossam Koth

Solution

16) $V_1 = 400 \text{ V}$ $f = 50 \text{ Hz}$

$I_m = 0.5 \text{ A}$ $\cos \phi_0 = 0.1 \text{ lag}$

At no load



$$I_m = I_0 \sin \phi_0 \quad \phi_0 = \cos^{-1}(0.1) = 84.26^\circ$$

$$I_0 = \frac{0.5}{\sin(84.26)} = 0.5025 \text{ A} \quad \#$$

17) $I_c = I_0 * \cos \phi_0$

$$\therefore I_c = 0.05 \text{ A} \quad \#$$

$$18) P_c = V * I_c = 400 * 0.05 = 20 \text{ W} \#$$

$$19) R_c = \frac{V}{I_c} = \frac{400}{0.05} = 8000 \Omega \#$$

$$20) X_m = \frac{V}{I_m} = \frac{400}{0.5} = 800 \Omega \#$$

$$21) 1\phi \text{ Transformer } (240/120) \quad f = 50 \text{ Hz}$$

$$S_{F.L} = 2 \text{ KVA}$$

S.C Test

$$V_{s.c} = 40 \text{ V}$$

$$I_{s.c} = 8 \text{ A}$$

$$P_{s.c} = 192 \text{ W}$$

$$PF = 0.6 \text{ lead} \& \text{ Transformer works at full-load}$$

$$V_2 = V_{load} = 120 \text{ V} \quad P_c = P_{n.L} = 40 \text{ W}$$

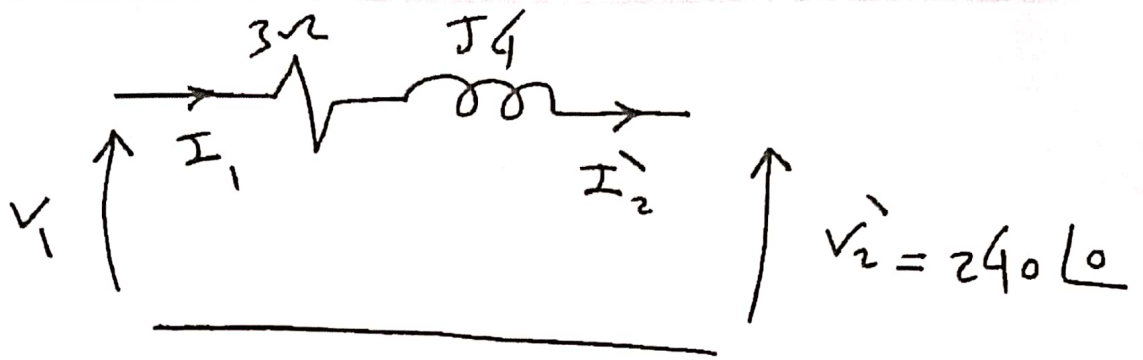
$$R_{eq} = \frac{192}{8^2} = 3 \Omega \#$$

$$22) Z = \frac{V_{s.c}}{I_{s.c}} = \frac{40}{8} = 5 \Omega$$

$$X_{eq} = \sqrt{Z^2 - R_{eq}^2} = 4 \Omega$$

$$4 = 2\pi * 50 * L_{eq} \quad \therefore L_{eq} = 0.0127 \text{ H} \#$$

23)



$$I_2 \simeq I_1 = \frac{2000}{240} \sqrt{+ \cos^{-1} 0.6}$$

$$I_2 \simeq I_1 = 8.33 \angle 53.13^\circ \text{ A} \#$$

$$\therefore V_1 = V_2 + I_2 (3 + j4)$$

$$\therefore V_1 = 231.812 \angle 9.93^\circ \text{ V} \#$$

$$24) \% \text{ V.R} = \frac{V_1 - V_2}{V_2} \times 100 \%$$

$$\% \text{ V.R} = \frac{231.81 - 240}{240} \times 100 = -3.4 \% \#$$

$$25) \eta_{F.L} = \frac{P_o}{P_o + P_c + P_{cu}} \times 100 \%$$

$$P_o = S \times P.F = 2000 \times 0.6 = 1200 \text{ W} \#$$

$$P_c = 40 \text{ W} \quad P_{cu} = |I_2|^2 \times R_{eq} = 208.17 \text{ W}$$

$$\therefore \eta_{F.L} = 82.86 \% \#$$

$$26) S_{\eta_{max}} = S_{f.L} * \sqrt{\frac{P_c}{P_{cu f.L}}}$$

$$S_{\eta_{max}} = 2000 * \sqrt{\frac{40}{208.17}} = 876.7 \text{ VA} \#$$

note If we want $\eta_{max} = \frac{S_{\eta_{max}} * PF}{S_{\eta_{max}} * PF + 2P_c}$

$$27) \begin{array}{l} \text{at } \left. \begin{array}{l} V_1 = 440 \text{ V} \\ F_1 = 50 \text{ Hz} \end{array} \right\} P_{C1} = 2500 \text{ W} \\ \text{at } \left. \begin{array}{l} V_2 = 220 \text{ V} \\ F_2 = 25 \text{ Hz} \end{array} \right\} P_{C2} = 850 \text{ W} \end{array}$$

$$P_c = P_e + P_h = K_e B_m^2 f^2 + K_h B_m f$$

$$2500 = K_e' (50)^2 + K_h' (50) \rightarrow (1)$$

$$850 = K_e' (25)^2 + K_h' (25) \rightarrow (2)$$

$$\text{As } \frac{V}{f} = \text{Const} \rightarrow B_m = \text{Const}$$

By solving (1) & (2)

$$K_e' = 0.64$$

$$K_h' = 18$$

$$P_e = 0.64 * (50)^2 = 1600 \text{ W} \#$$

$$28) P_h = 18 \times 50 = 900 \text{ W} \quad \#$$

$$\text{or } P_h = P_c - P_e = 2500 - 1600 = 900 \text{ W} \quad \#$$

$$29) P_h = K'_h \times f$$

$$P_h = 18 \times 25 = 450 \text{ W} \quad \#$$

$$30) \text{ If } \frac{v_1}{v_2} = \frac{f_1}{f_2} = \text{Const}$$

$$\therefore B_{\max} = \text{Const} \quad \#$$

VIP

Two winding Transformer

$$\%Z = 0.01 + j0.02$$

The Transformer is operating at full load
and rated volt with 0.8 pf lag

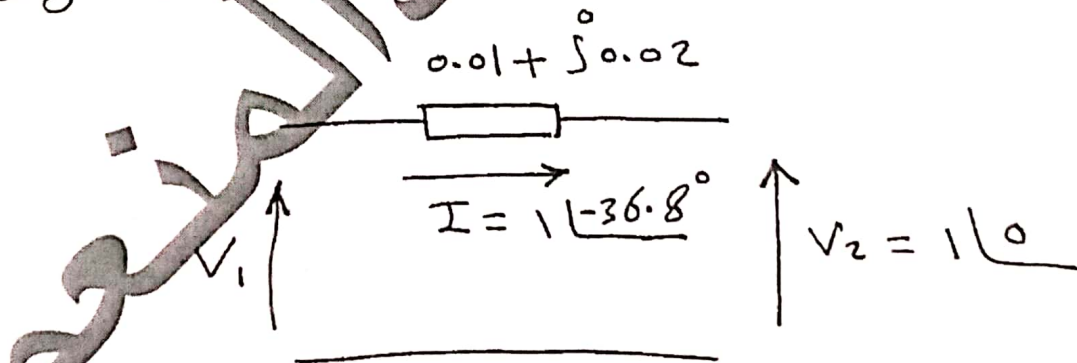
Find 1) Voltage regulation

2) η_{max}

if core loss is 90% of full load copper loss

Soln

1) By using Per unit values



$$\bar{V}_1 = \bar{V}_2 + \bar{I} \bar{Z} \quad \therefore V_1 = 1.02 \angle 0.563^\circ$$

$$\% V.R = \frac{1.02 - 1}{1} * 100 \%$$

$$\% V.R = 2 \% \quad \#$$

$$2) \quad S_{\eta_{max}} = S_{P_{f.L}}^* \sqrt{\frac{P_C}{P_{Cu f.L}}} = \sqrt{0.9}$$

$$\therefore S_{\eta_{max}} = 0.94868 \text{ Pu}$$

$$\eta_{max} = \frac{0.94868 * 0.8}{(0.94868 * 0.8) + 2 * P_C} * 100\%$$

$$P_{Cu f.L} = |I|^2 * R_{eq} = (1)^2 * 0.01 = 0.01 \text{ Pu}$$

$$P_C = 0.9 P_{Cu f.L} = 0.009 \text{ Pu}$$

$$\therefore \eta_{max} = 97.68 \% \quad \#$$

Final Model

Q.1

A 4kVA, 400/200V, 50Hz, single phase transformer gave the following test figures:

No load test : 400V, 0.5A, 60W

Short circuit test (HV data) : 9V, 6A, 21W

Determine:

(a) The Core resistance and magnetizing reactance.

(b) Equivalent resistance and reactance.

If the transformer works at full load 0.8 pf lagging with load voltage 200V,

(c) Determine the voltage regulation.

(d) The efficiency.

(e) The efficiency at 75%full load unity power factor.

Q.2

For the 100/50V single phase-transformer shown in Fig. 1, the primary and secondary windings are connected in series, where the winding dot notations are missing. With an applied voltage of 100V, the voltage across each winding is measured using an AC voltmeter.

6. If the voltmeter reading V_1 is 200V, then the dot notation for the primary and secondary windings should likely be placed on terminals

- a. 1 and 2 b. 1 and 3 ~~c. 1 and 4~~ d. 2 and 4

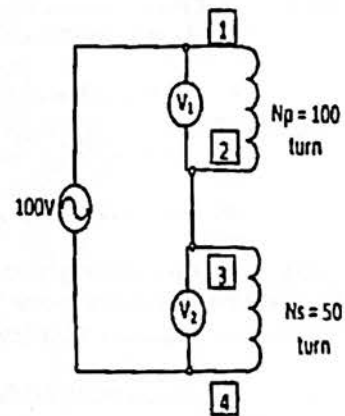


Fig. 1

7. If the dot notations for the primary and secondary winding are placed on terminal 2 and 4, respectively, the AC voltmeter readings V_1 and V_2 will likely be

- a. $V_1 = 150V, V_2 = 50V$ ~~b. $V_1 = 67V, V_2 = 33V$~~ c. $V_1 = 200V, V_2 = 100V$ d. $V_1 = 50V, V_2 = 50V$

Q.3

- Back To Back test on transformer is used to determine:
 - a-max sc capacity
 - ~~b~~- max temp rise
 - c-max operating volt
 - d-all of above

- 26) For parallel operation of two transformers, they should have
- a. The same turns ratio and voltage rating
 - b. The same % Impedance, and X/R ratio
 - c. The same polarity
 - ~~d. All of the above~~

- 27) If two transformers connected in parallel with different X/R ratio
- a. Each transformer will share different power factor
 - b. Each transformer will share the same load's power factor
 - c. One transformer could be over loaded
 - ~~d. (a), and (c)~~

• Questions (28→30)

• Two transformers have a turns ratio equivalent to their 2400 - 240 volta

The data of each one is listed in the following table:

A	B
75-kVA	200-kVA
$Z_{Aeq} = 2.7342 \angle 62.571^\circ \Omega$	$Z_{Beq} = 1.2031 \angle 74.733^\circ \Omega$

- 28) The percent of the total bank-current drawn by each transformer is
- a. $I_A = 0.698 I_{bank}$, $I_B = 0.307 I_{bank}$

b. $I_A = 0.5 I_{bank}$, $I_B = 0.5 I_{bank}$

~~c. $I_A = 0.307 I_{bank}$, $I_B = 0.698 I_{bank}$~~

d. $I_A = 0.75 I_{bank}$, $I_B = 0.25 I_{bank}$

- 29) The maximum load that can be handled by the bank without over load of the transformers

a. $S_{bank} = 520\text{-kVA}$

~~b. $S_{bank} = 244\text{-kVA}$~~

c. $S_{bank} = 275\text{-kVA}$

d. $S_{bank} = 175\text{-kVA}$

- 30) The % voltage regulation, if the maximum load is unity power Factor
- a. 1.8 %
 - b. -3.2 %
 - c. 1.2 %

2.4

- State two definitions different for percentage impedance?

Q.5

Connecting two 2000 kVA transformers in parallel, one with 5.75% impedance and the other with 4% impedance, each with the same turn ratios, connected to a common 3500 kVA load. What is the loading on each transformer?

Q.6

A load of 500 A, at 0.8 power (lagging), at a terminal voltage of 400 V is supplied by two transformers are connected in parallel. The equivalent impedances of the two transformers referred to the secondary sides are $(2 + j3)$ ohm and $(2.5 + j5)$ ohm respectively. Calculate the current and kVA supplied by each transformer and the power factor at which they operate.

Q.7

A 480/120v, 5KVA two winding transformer is to be used as an auto-transformer to supply power at 480v from 600v source.

Determine:

- a- KVA of the autotransformer
- b- $S_{\text{conductive}}$
- c- $S_{\text{inductive}}$
- d- If the ampere turn of two winding transformer was 100 A.T determine the saving in the copper when using autotransformer.

Q.8

A 240/208 V, 6 KVA, 50 Hz, three-phase, delta-star transformer has the following test results:

- With secondary side open-circuited the primary side data are: 240 V, 4 A, 270 W.
- With secondary side short-circuited the primary side data are: 40 V, 14 A, 288 W.

Determine:

- i. The transformer per phase parameters R_e , X_m , R_{eq1} and X_{eq1} .
- ii. The primary voltage at full load, 0.8 power factor leading and 208 V on the secondary terminals.
- iii. The voltage regulation and transformer efficiency for the loading in part "ii".

Q.9

For the three-phase transformer connection shown in Fig. 3 having a winding turns ratio a ,

31. The corresponding vector group will be

- a. $\Delta y1$ b. $\Delta y3$ c. $\Delta y5$ d. $\Delta y7$

32. The corresponding voltage ratio will be

- a. a b. $a/\sqrt{3}$ c. $\sqrt{3}a$ d. $a/3$

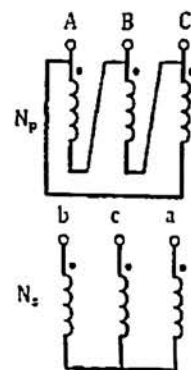


Fig. 3

Q.10

51. A V-V connected transformer can be connected in parallel to another transformer having a connection

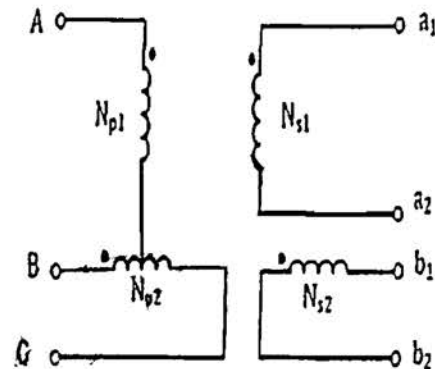
- a. Δ -Y b. Y- Δ c. Y-V ~~d. Δ - Δ~~

52. If one of the transformers is removed from the bank of delta-delta connected three-phase transformer, then the maximum load will be ---- of the total rated power

- a. 33% ~~b. 58%~~ c. 66% d. 100%

64. In order to obtain a balanced two-phase output then

- a. $N_{s1} = N_{s2}, N_{p1} = N_{p2}$ c. $N_{s1} = 0.87N_{s2}, N_{p1} = N_{p2}$
b. $N_{s1} = N_{s2}, N_{p2} = 0.87N_{p1}$ ~~d. $N_{s1} = N_{s2}, N_{p1} = 0.87N_{p2}$~~



Q.11

Two single phase transformers are V-connected supplying a balanced 3-ph Y load of 100 kVA at line voltage of 220 V, 0.8 pf lag.

- (a) Draw the vector diagram.
(b) find the pf of each transf.
(c) find the power supplied by each transf.

Q.1

a)

$$\cos \phi_0 = \frac{P_0}{V_0 I_0} = \frac{60}{400 \times 0.5} = 0.3$$

1 percent
the

$$I_c = I_0 \times \cos \phi_0 = 0.15 A$$

$$I_m = I_0 \times \sin \phi_0 = 0.477 A$$

$$R_c = \frac{V_0}{I_c} = 2666.67 \Omega$$

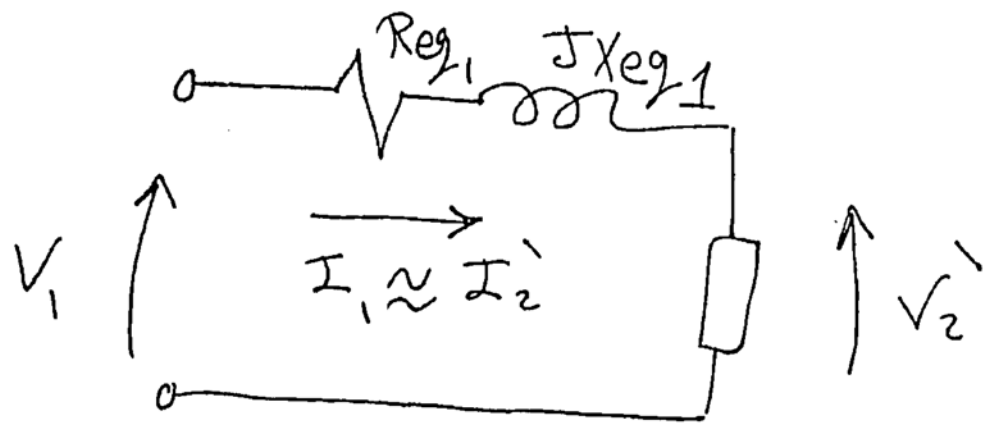
$$X_m = \frac{V_0}{I_m} = 838.6 \Omega$$

b)

$$R_{eq1} = \frac{P_{sc}}{I_{sc}^2} = 0.58 \Omega$$

$$Z_{eq1} = \frac{V_{sc}}{I_{sc}} = 1.5 \Omega$$

$$X_{eq1} = \sqrt{Z_{eq1}^2 - R_{eq1}^2} = 1.38 \Omega$$



$$V_2' = 200 \times \frac{400}{200} = 400 \angle 0^\circ \text{ V}$$

$$I_1 = \frac{4000}{400} \angle -\cos^{-1} 0.8 \text{ A}$$

$$\therefore I_1 = 10 \angle -36.86^\circ \text{ A}$$

$$V_1 = V_2' + I_2' \times (R_{eq1} + jX_{eq1})$$

$$V_1 = 400 \angle 0^\circ + (10 \angle -36.86^\circ)(0.58 + j1.38)$$

$$\therefore V_1 = 412.98 \angle 1.05^\circ \text{ V}$$

$$\% \text{ V.R} = \frac{|V_1| - |V_2'|}{|V_2'|} \times 100 \%$$

$$\therefore \% \text{ V.R} = 3.24 \%$$



$$d) \eta = \frac{P_o}{P_o + P_c + P_{cu}}$$

$$\eta = \frac{4000 \times 0.8}{(4000 \times 0.8) + 60 + (10)^2 \times 0.58}$$

$$\therefore \eta = 96.44 \% \quad \#$$

$$e) \eta_x = \frac{x P_o}{x P_o + P_c + x^2 P_{cu}}$$

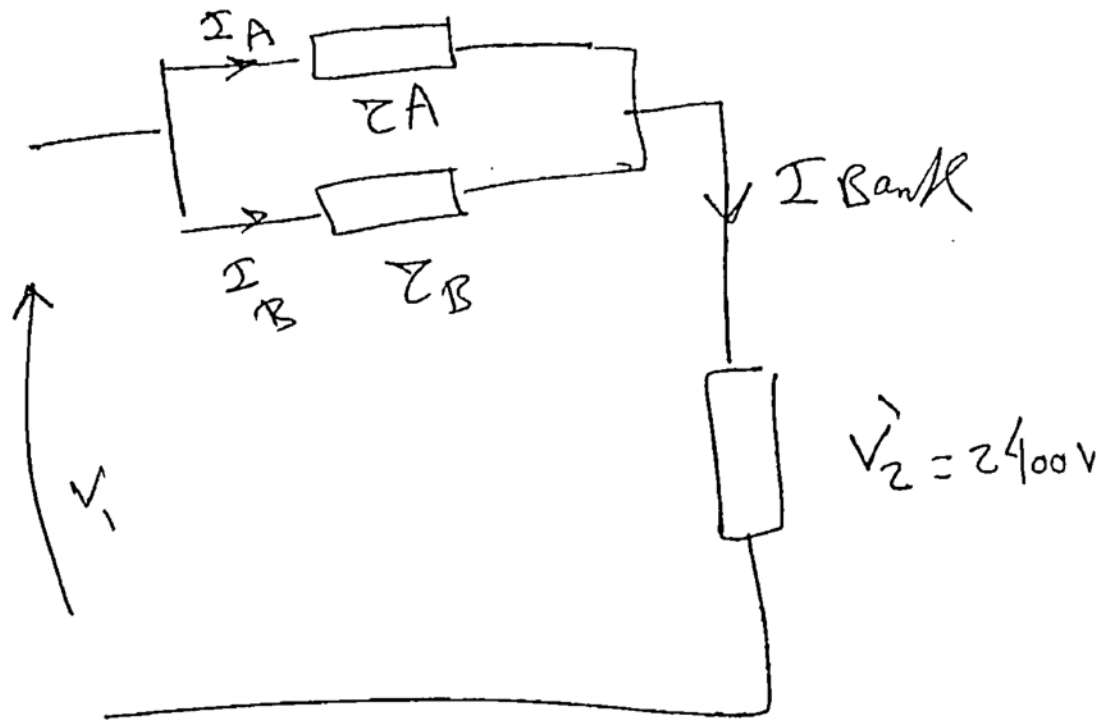
$$\eta_{75\%} = \frac{0.75 \times 4000 \times 1}{(0.75 \times 4000 \times 1) + (60) + (10)^2 \times 0.58 \times (0.75)}$$

$$\eta_{75\%} = 97 \%$$

Note * $\eta_{max} = \frac{S_{\eta_{max}} \cdot Pf}{S_{\eta_{max}} Pf + 2 P_c}$

$$* S_{\eta_{max}} = S_{f.L} \times \sqrt{\frac{P_c}{P_{cu} f.L}}$$

28)



$$I_A = I_{Bank} * \left| \frac{Z_B}{Z_A + Z_B} \right|$$

$$\therefore I_A = 0.307 I_{Bank} \quad \#$$

$$I_B = I_{Bank} * \left| \frac{Z_A}{Z_A + Z_B} \right|$$

$$\therefore I_B = 0.698 I_{Bank}$$

#

$$29) \quad I_{A \text{ rated}} = \frac{S_A}{V_1} = 31.25 \text{ A}$$

$$I_{B \text{ rated}} = \frac{S_B}{V_1} = 83.33 \text{ A}$$

$$\underline{\underline{\text{If}}} \quad I_A = I_{A \text{ rated}} = 31.25 \text{ A}$$

$$\therefore I_{\text{bank}} = 101.7915 \text{ A}$$

$$\therefore I_B = 71.05 \text{ A}$$

$$I_B < I_{B \text{ rated}}$$

$\therefore$ No over load

$$S_{\text{bank}} = I_{\text{bank}} \times V_1$$

$$S_{\text{bank}} = 244 \text{ KVA}$$

✓ KVL

$$\leadsto V_1 = V_2 + I_{\text{bank}} \times Z_{eq}$$

$$V_1 = 2400 \angle 0^\circ + (101.79 \angle 0^\circ) (0.8 \angle 71.024^\circ)$$

$$\therefore V_1 = 2427.701 \angle 3.8^\circ$$

$$V.R = \frac{|V_1| - |V_2|}{|V_2|} \times 100 \%$$

$$V.R = 1.154 \%$$

$$V.R \approx 1.2 \%$$

Q.5

$$\text{Tr. A loading} = \frac{S_A / \%Z_A}{\frac{S_A}{\%Z_A} + \frac{S_B}{\%Z_B}} * S_L$$

$$\text{Tr. A loading} = \frac{\frac{2000}{5.75}}{\frac{2000}{5.75} + \frac{2000}{4}} * 3500$$

$$\text{Tr. A loading} = 1436 \text{ KVA}$$

→ No overload

$$\text{Tr. B loading} = \frac{\frac{2000}{4}}{\frac{2000}{5.75} + \frac{2000}{4}} * 3500$$

$$\text{Tr. B loading} = 2064 \text{ KVA}$$

→ overloaded

1.6

$$I_{Bank} = 500 \angle -\cos^{-1} 0.8 = 500 \angle -36.87^\circ A$$

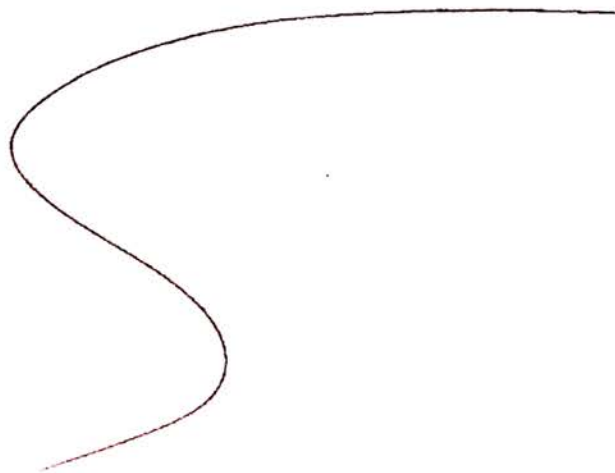
$$I_1 = I_{Bank} * \frac{Z_2}{Z_1 + Z_2} = 304.5 \angle -34.08^\circ A$$

$$I_2 = I_{Bank} - I_1 = 196.41 \angle -41.2^\circ A$$

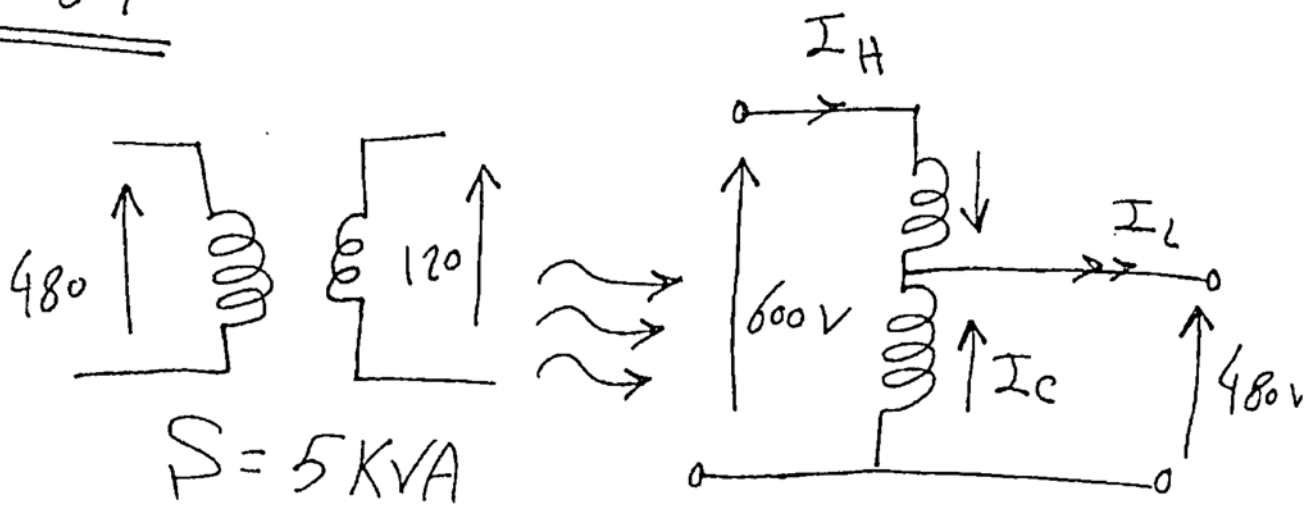
$$Pf_1 = \cos(34.08) = 0.8189 \text{ lag}$$

$$Pf_2 = \cos(41.2) = 0.755 \text{ lag}$$

$$S_{bank} = 500 * 400 = 200 \text{ KVA}$$



Q.7



a)

$$I_H = \frac{5000}{120} = 41.67 \text{ A}$$

$$I_C = \frac{5000}{480} = 10.42 \text{ A}$$

$$I_L = I_H + I_C = 52.09 \text{ A}$$

$$S_{\text{Auto}} = I_H \times V_H = 41.67 \times 600$$

$$\therefore S_{\text{Auto}} = 25 \text{ KVA} \quad \#$$

$$b) \quad S_{\text{conductive}} = K \times S_{\text{Auto}}$$

$$K = \frac{V_2}{V_1} = \frac{480}{600} = 0.8$$

$$\therefore S_{\text{conductive}} = 0.8 \times 25 = 20 \text{ KVA}$$

$$c) S_{\text{inductive}} = S_{\text{Auto}} - S_{\text{conductive}}$$

$$\therefore S_{\text{inductive}} = 25 - 20 = 5 \text{ KVA}$$

$$d) W_a = (1 - k) * W_o$$

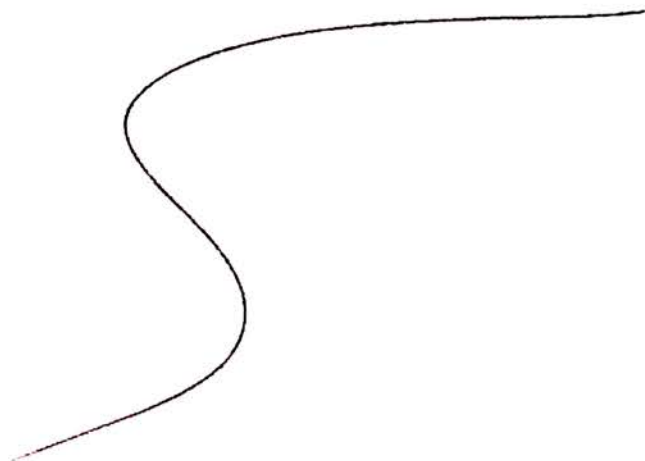
$$W_a = (1 - 0.8) * 100$$

$$\therefore W_a = 20 \text{ A.T}$$

$$\therefore \text{Saving} = W_o - W_a$$

$$\therefore \text{Saving} = 100 - 20 = 80 \text{ A.T}$$

#



Q.8

$$a = \frac{V_{ph1}}{V_{ph2}} = \frac{240}{208/\sqrt{3}} = 2$$

$$S_{f.L} = 6000 \text{ VA}$$

$$I_2' \approx I_1 = \frac{6000}{3 \times 240} = 8.333 \text{ A}$$

i) (Per Phase values)

$$V_0 = 240 \text{ V}, \quad I_0 = \frac{4}{\sqrt{3}} \text{ A}, \quad P_0 = \frac{270}{3} = 90 \text{ W}$$

$$\phi_0 = \cos^{-1} \left(\frac{P_0}{V_0 I_0} \right) = 80.65^\circ$$

$$I_c = I_0 \cos \phi_0 = 0.375 \text{ A}$$

$$I_m = I_0 \sin \phi_0 = 2.279 \text{ A}$$

$$R_c = \frac{V_0}{I_c} = 639.8 \Omega$$

$$X_m = \frac{V_0}{I_m} = 105.3 \Omega$$

(Per Phase values)

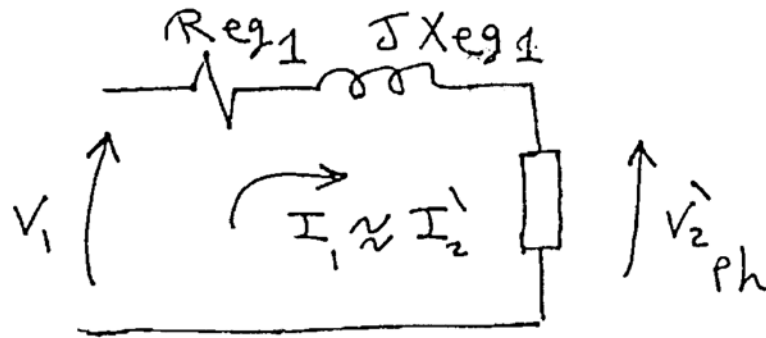
$$V_{s.c} = 40 \text{ V}, I_{s.c} = \frac{14}{\sqrt{3}} \text{ A}, P_{sc} = \frac{288}{3} = 96 \text{ W}$$

$$R_{eq1} = \frac{P_{sc}}{I_{sc}^2} = 1.47 \Omega$$

$$Z_{eq1} = \frac{V_{sc}}{I_{sc}} = 4.95 \Omega$$

$$X_{eq1} = \sqrt{Z_{eq1}^2 - R_{eq1}^2} = 4.73 \Omega$$

(ii)



$$V_{2_{ph}} = \frac{208}{\sqrt{3}} * a = 240 \text{ V} \neq$$

$$I_2 = 8.333 \angle 36.87^\circ \rightarrow \text{lead PF}$$

$$V_1 = V_{2_{ph}} + I_2 (R_{eq1} + jX_{eq1})$$

$$\therefore V_1 = 229.47 \angle 9.75^\circ$$

$$iii) V.R = \frac{V_1 - V_2}{V_2} * 100 \%$$

$$V.R = \frac{(229.47) - (240)}{(240)} * 100 \%$$

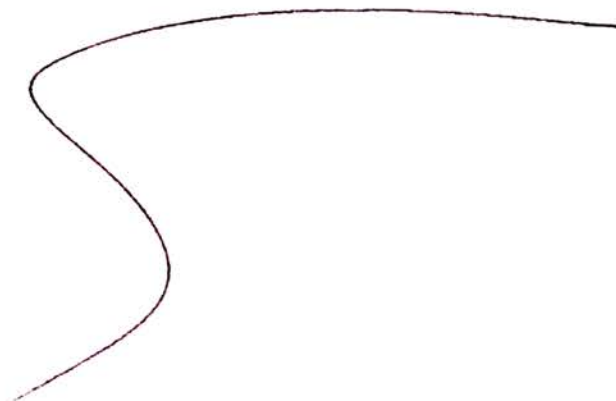
$$\therefore V.R = -4.39 \%$$

$$\eta = \frac{P_o}{P_o + P_c + P_{cu}} * 100 \%$$

$\sim 3 I_a^2 R_{eq1}$

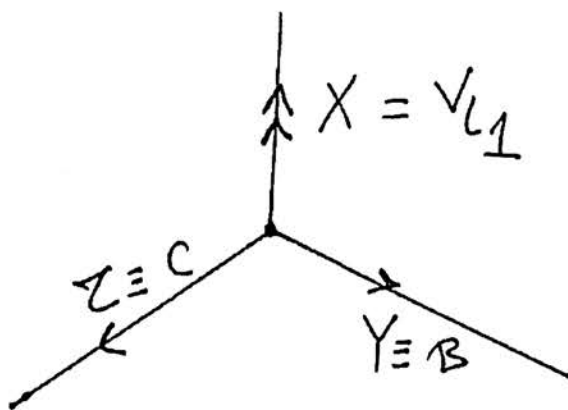
$$\eta = \frac{(6000 * 0.8) * 100\%}{(6000 * 0.8) + (270) + 3 * (8.333)^2 * 1.47}$$

$$\therefore \eta = 89.28 \%$$

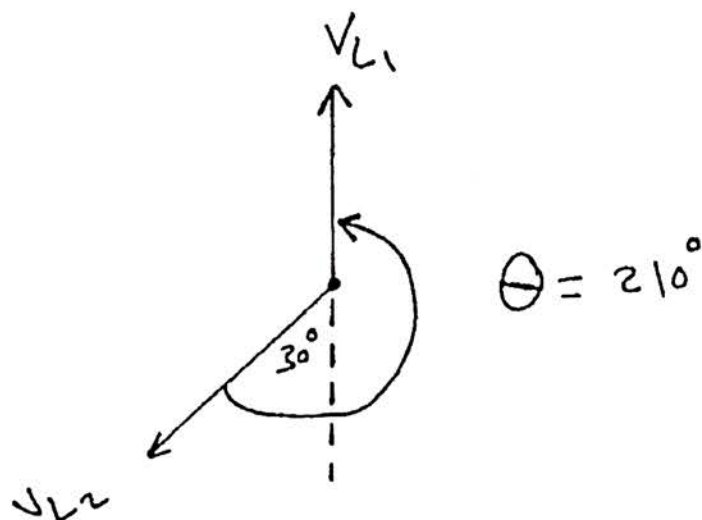
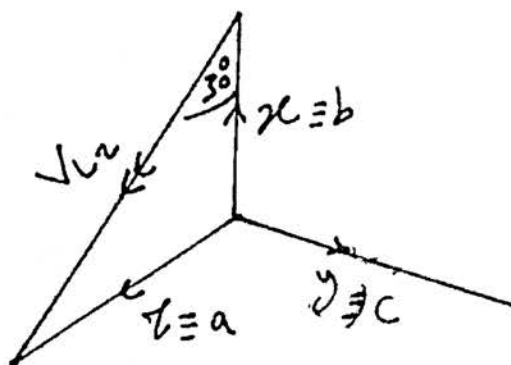


31)

$$V_{AB} = V_A$$



$$V_{AB} = V_A - V_B$$

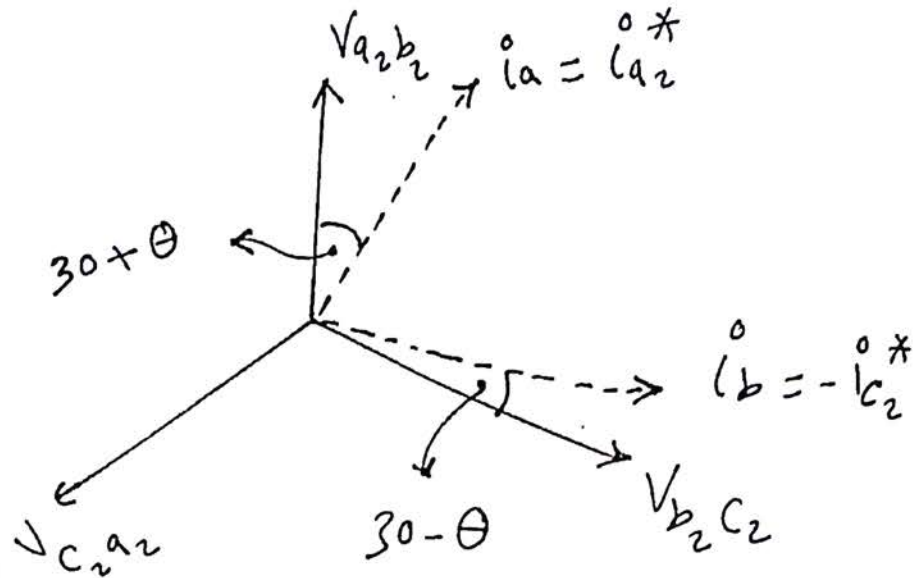


V.G $\rightarrow$ Dy 7

32) $a/\sqrt{3}$

Q.11

a)



$$b) \theta = \cos^{-1} \text{PF} = \cos^{-1}(0.8) = 36.87^\circ$$

$$\text{PF of TR.1} = \cos(30 + \theta) = 0.393 \#$$

$$\text{PF of TR.2} = \cos(30 - \theta) = 0.993 \#$$

$$c) P_{\text{TR.1}} = V_{\text{ph}} I_{\text{ph}} \text{PF}_1$$

$$P_{\text{TR.1}} = 220 \times \frac{100000}{\sqrt{3} \times 220} \times 0.393$$

$$P_{\text{TR.1}} = 22.68 \text{ KW} \#$$

$$P_{\text{TR.2}} = 220 \times \frac{100000}{\sqrt{3} \times 220} \times 0.993$$

$$P_{\text{TR.2}} = 57.32 \text{ KW} \#$$

1 Φ TR $\rightarrow$ 100/50

* if $V_1 = 200V$

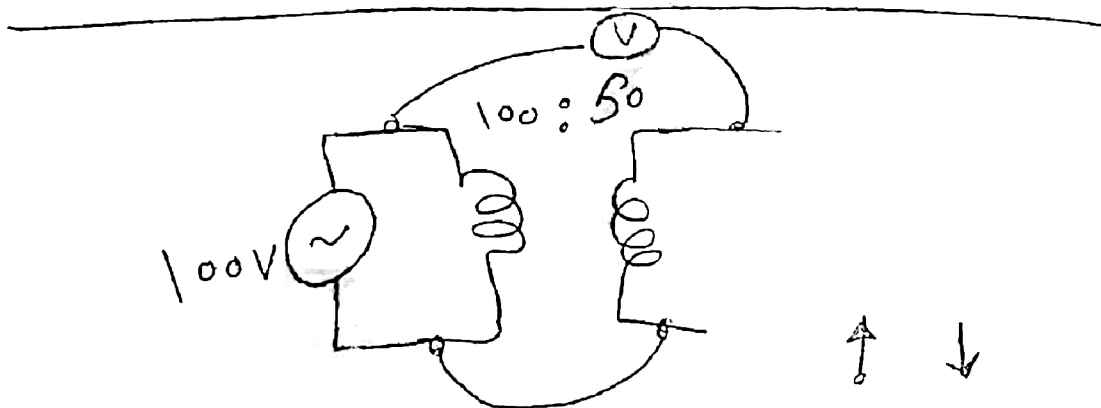
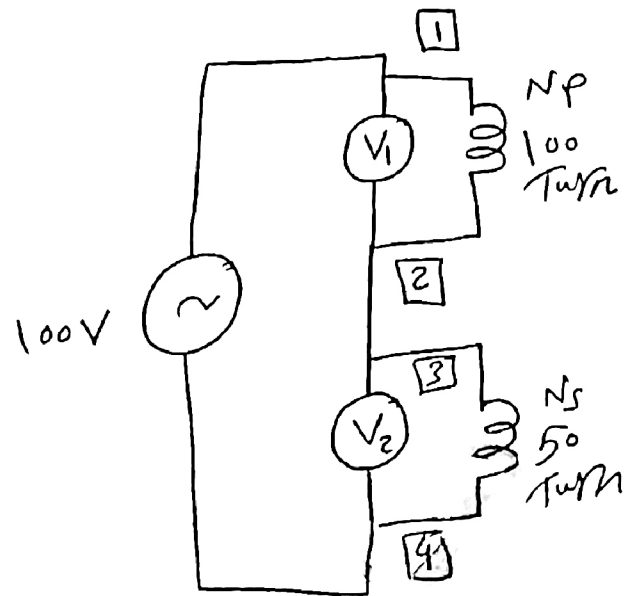
dot notation

$\rightarrow$ 1 & 4

* If dot $\rightarrow$ 2
 $\rightarrow$ 4

$V_1 = 150, 67, 200, 50$

$V_2 = 50, 33, 100, 50$



$\checkmark \rightarrow 150V \rightarrow$ Additive Polarity
 $\checkmark \rightarrow 50V \rightarrow$ subtractive Polarity
 $\uparrow \uparrow$

EX

during back-back test

$$S = 250 \text{ kVA} \quad W_1 = 5 \text{ kW}$$

$$W_2 = 7.5 \text{ kW}$$

$$\text{PF} = 0.8 \text{ lag}$$

at 75% load find η for each transformer

Soln

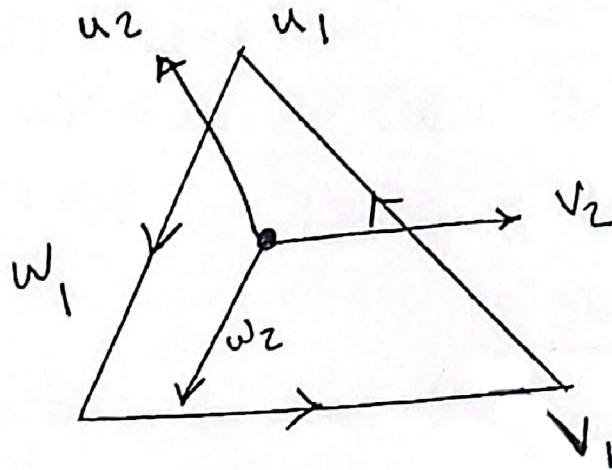
$$P_c = \frac{5}{2} = 2.5 \text{ kW}$$

$$P_{Cu_{FL}} = \frac{7.5}{2} = 3.75 \text{ kW}$$

because wattmeter calculates for two transformers

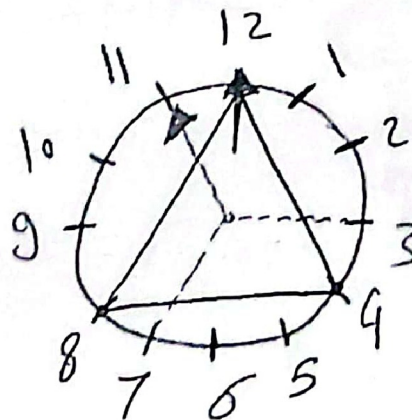
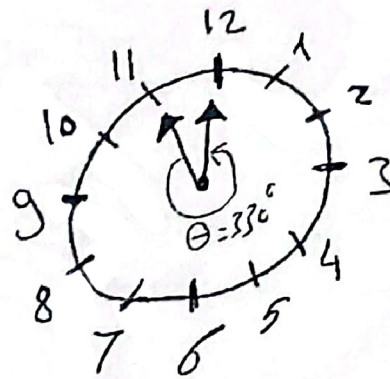
$$\eta = \frac{S \times \text{PF} \times X}{S \times \text{PF} \times X + P_c + X^2 P_{Cu_{FL}}} \times 100$$

$$\therefore \eta = 97\%$$



Find vector Group

clock notation



V.G

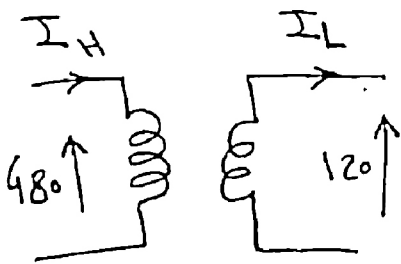
Dy 11

EX-3

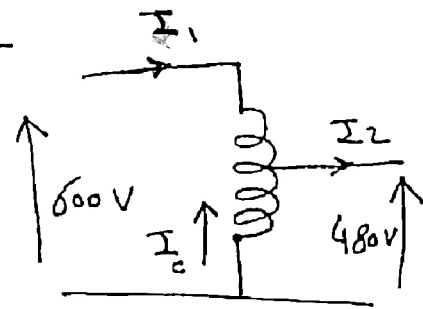
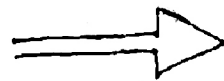
Two winding Transformer 480/120 V
5 KVA used as Auto Transformer
to supply power at 480 V from 600 V

Calculate :

- KVA of Auto Transformer
- saving in G.P.P. weight
if $w_0 = 100 \text{ A.T}$



Solution



$$I_1 = \frac{5 \times 10^3}{120} = 41.67 \text{ A}$$

$$I_c = \frac{5 \times 10^3}{480} = 10.417 \text{ A}$$

$$I_2 = I_c + I_1 = 52.087 \text{ A}$$

$$S_{\text{auto}} = I_1 V_1 = I_2 V_2$$

$$* K = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2} \quad * S_{\text{cond}} = S_{\text{auto}} * K$$

$$S_{anTo} = 41.67 \times 600 = 25 \text{ KVA}$$

#

$$b) W_{anTo} = W_0 (1 - K)$$

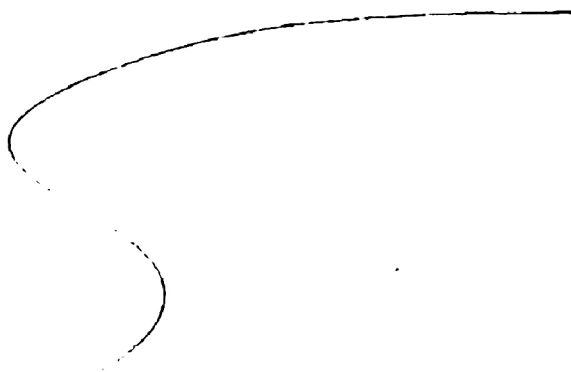
$$K = \frac{V_2}{V_1} = \frac{480}{600} = 0.8$$

$$\therefore W_a = 100 (1 - 0.8) = 20 \text{ A.T}$$

$$\therefore \text{Saving of Gller} = W_0 - W_a$$

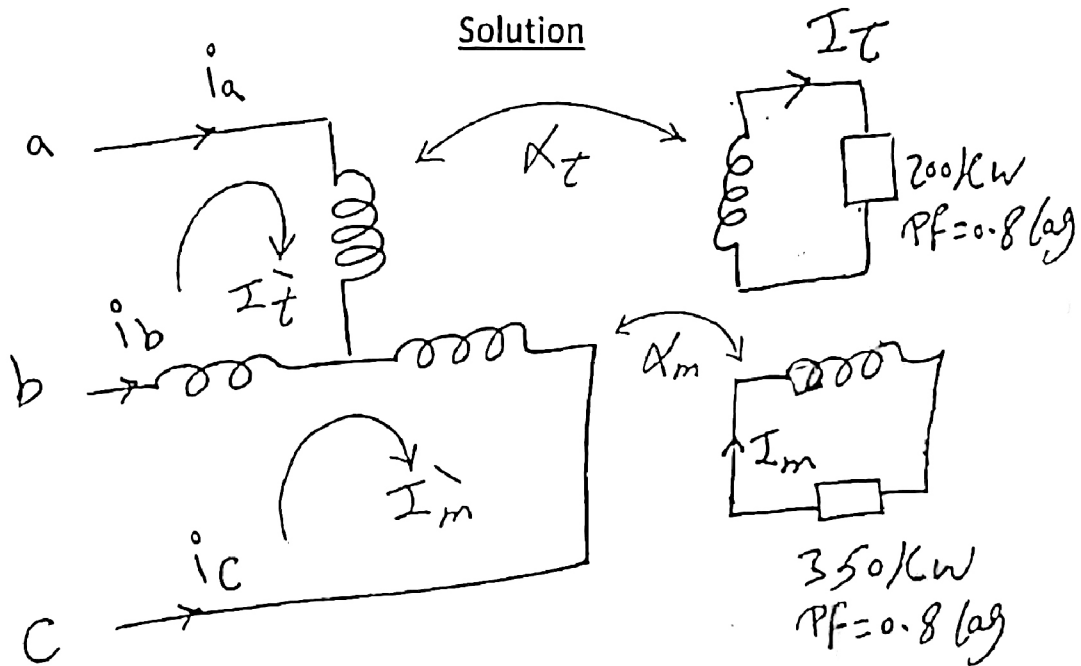
$$\text{Saving of Gller} = 100 - 20 = 80 \text{ A.T}$$

#



Example 33.22. Two furnaces are supplied with 1-phase current at 50 V from a 3-phase, 4.6 kV system by means of two 1-phase, Scott-connected transformers with similar secondary windings. When the load on the main transformer is 350 kW and that on the other transformer is 200 kW at 0.8 p.f. lagging, what will be the current in each 3-phase line? Neglect phase displacement and losses in transformers.
(Electrical Machinery-II, Bangalore Univ. 1991)

Solution



$$3\phi \text{ line volt} = 4.6 \text{ KV}$$

$$1\phi \text{ volt} = 50 \text{ V}$$

$$\alpha_m = \frac{4600}{50} = 92$$

$$\alpha_t = \frac{\sqrt{3}}{2} \alpha_m = 46\sqrt{3}$$

$$I_m = \frac{350 \times 10^3}{50 \times 0.8} \angle -\cos^{-1} 0.8$$

$$\therefore I_m = 8750 \angle -36.869^\circ \quad \#$$

$$I_t = \frac{200 \times 10^3}{50 \times 0.8} \angle 90 - \cos^{-1} 0.8$$

$$\therefore I_t = 5000 \angle 53.13^\circ \quad \#$$

$$I_m' = \frac{I_m}{\alpha_m} = 95.11 \angle -36.869^\circ$$

$$I_t' = \frac{I_t}{\alpha_t} = 62.755 \angle 53.13^\circ$$

$$I_a = I_t' = 62.755 \angle 53.13^\circ \quad \#$$

$$I_b = I_m' - \frac{I_t'}{2} = 100.15 \angle -55.127^\circ \quad \#$$

$$I_c = \square \square I_m' - \frac{I_t'}{2} = 100.15 \angle 161.389^\circ \quad \#$$

Example 33.13. A $\Delta - \Delta$ bank consisting of three 20-kVA, 2300/230-V transformers supplies a load of 40 kVA. If one transformer is removed, find for the resulting V-V connection

- (i) kVA load carried by each transformer
- (ii) per cent of rated load carried by each transformer
- (iii) total kVA rating of the V-V bank
- (iv) ratio of the V-V bank to $\Delta - \Delta$ bank transformer ratings.
- (v) per cent increase in load on each transformer when bank is converted into V-V bank.

Solution

$$(i) \Rightarrow \frac{40 \text{ kVA}}{\sqrt{3}} = 23.1 \text{ kVA}$$

$$(ii) \Rightarrow \frac{23.1}{20} \times 100 = 115.5 \%$$

$$(iii) \Rightarrow 2 \times 20 \times 0.866 = 34.64 \text{ kVA}$$

$$(iv) \Rightarrow \frac{34.64}{3 \times 20} \times 100 = 57.7 \%$$

$$(v) \Rightarrow \text{load supplied by } \Delta - \Delta \text{ bank} \\ 40/3 = 13.33 \text{ kVA}$$

$$\frac{23.1}{13.3} \times 100 = 173.2 \%$$

